Then,

$$
\begin{aligned}
P(k+1) &= P(k) + \cos(\alpha + k\beta) \\
&= \frac{\cos\left(\alpha + \frac{(k-1)\beta}{2}\right)\sin\left(\frac{k\beta}{2}\right)}{\sin(\frac{\beta}{2})} + \cos(\alpha + k\beta) \\
&= \frac{1}{\sin(\frac{\beta}{2})}\left[\cos\left(\alpha + \frac{(k-1)\beta}{2}\right)\sin\left(\frac{k\beta}{2}\right) + \cos(\alpha + k\beta)\sin\left(\frac{\beta}{2}\right)\right] \\
&= \frac{1}{\sin(\frac{\beta}{2})}\left[\cos\left(\left(\alpha + \frac{k\beta}{2}\right) - \frac{\beta}{2}\right)\sin\left(\frac{k\beta}{2}\right) + cos(\alpha + k\beta)\sin\left(\frac{\beta}{2}\right)\right] \\
&= \frac{1}{\sin\left(\frac{\beta}{2}\right)}\left[\left(\cos\left(\alpha + \frac{k\beta}{2}\right)\cos\left(\frac{\beta}{2}\right)\right.\right. \\
&\quad \left.\left. + \sin\left(\alpha + \frac{k\beta}{2}\right)\sin\left(\frac{\beta}{2}\right)\right)\sin\left(\frac{k\beta}{2}\right) + \cos(\alpha + k\beta)\sin\left(\frac{\beta}{2}\right)\right] \\
&= \frac{1}{\sin\left(\frac{\beta}{2}\right)}\left[\cos\left(\alpha + \frac{k\beta}{2}\right)\cos\left(\frac{\beta}{2}\right)\sin\left(\frac{k\beta}{2}\right)\right. \\
&\quad \left. + \sin\frac{\beta}{2}\left(\sin\left(\alpha + \frac{k\beta}{2}\right)\sin\left(\frac{k\beta}{2}\right) + \cos(\alpha + k\beta)\right)\right] \\
&= \frac{1}{\sin\left(\frac{\beta}{2}\right)}\left[\cos\left(\alpha + \frac{k\beta}{2}\right)\cos\left(\frac{\beta}{2}\right)\sin\left(\frac{k\beta}{2}\right)\right. \\
&\quad \left. + \frac{\sin\frac{\beta}{2}}{2}\left(2\sin\left(\alpha + \frac{k\beta}{2}\right)\sin\left(\frac{k\beta}{2}\right) + 2\cos(\alpha + k\beta)\right)\right] \\
&= \frac{1}{\sin\left(\frac{\beta}{2}\right)}\left[\cos\left(\alpha + \frac{k\beta}{2}\right)\cos\left(\frac{\beta}{2}\right)\sin\left(\frac{k\beta}{2}\right)\right. \\
&\quad \left. + \frac{\sin\frac{\beta}{2}}{2}\left(\cos\alpha - \cos\left(\alpha + k\beta\right) + 2\cos(\alpha + k\beta)\right)\right] \\
&= \frac{1}{\sin\left(\frac{\beta}{2}\right)}\left[\cos\left(\alpha + \frac{k\beta}{2}\right)\cos\left(\frac{\beta}{2}\right)\sin\left(\frac{k\beta}{2}\right) + \frac{\sin\frac{\beta}{2}}{2}\left(\cos\alpha + \cos(\alpha + k\beta)\right)\right] \\
&= \frac{1}{\sin\left(\frac{\beta}{2}\right)}\left[\cos\left(\alpha + \frac{k\beta}{2}\right)\cos\left(\frac{\beta}{2}\right)\sin\left(\frac{k\beta}{2}\right) + \frac{\sin\frac{\beta}{2}}{2}\left(2\cos\left(\alpha + \frac{k\beta}{2}\right)\cos\left(\frac{-k\beta}{2}\right)\right)\right] \\
&= \frac{\cos\left(\alpha + \frac{k\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}\left[\sin\left(\frac{k\beta}{2}\right)\cos\left(\frac{\beta}{2}\right) + \sin\frac{\beta}{2}\cos\left(\frac{k\beta}{2}\right)\right] \\
&= \frac{\cos\left(\alpha + \frac{k\beta}{2}\right)\sin\left(\frac{(k+1)\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}
\end{aligned}
$$

That is,

$$\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \cdots + \cos(\alpha+(k-1)\beta) + \cos(\alpha+k\beta)$$

$$= \cos\left(\alpha + \frac{k\beta}{2}\right) \times \frac{\sin\left(\frac{(k+1)\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)}$$